

AeroPassport

Project Report

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Project Abstract

Aerospace hardware requires complete lifecycle traceability, including supplier details, batch information, inspection history, subsystem integration, and mission events. This project models that traceability problem in a lightweight and practical way using MongoDB.

AeroPassport manages flight hardware data including component information, suppliers, batch records, certifications, inspections, module assignments, and lifecycle events. The application is implemented as a console-based backend with REST APIs tested through Postman and Swagger. The project demonstrates MongoDB's document-oriented storage, embedded and referenced documents, relationship mapping, and CRUD capabilities.

Project Objective

The objective of AeroPassport is to demonstrate effective MongoDB usage through:

- Database connectivity using Spring Boot.
 - Console-based execution.
 - REST API validation using Postman and Swagger.
 - CRUD operations on core entities.
 - Insert, update, delete, and collection drop operations.
 - Relationship mapping using one-to-one, one-to-many, many-to-one, and many-to-many models.
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Key Features

- Flight Hardware Registration
 - Digital Part Passport Generation
 - Embedded Inspection and Lifecycle Tracking
 - Supplier Management
 - Module Management
 - Hardware Genealogy Report
 - Administrative Statistics Dashboard
 - Automatic Event Logging
 - MongoDB Aggregation Pipeline
 - MongoDB Indexing
 - REST API-based CRUD Operations
-

Technology Stack

Component	Technology
Language	Java 21
Framework	Spring Boot
Database	MongoDB Community Server
Data Access	Spring Data MongoDB, MongoTemplate
API Testing	Swagger UI, Postman
Build Tool	Maven Wrapper
IDE	IntelliJ IDEA
Database Tool	MongoDB Compass
Testing Tools	Postman, Swagger UI
Version Control	Git

Layered Architecture

Presentation Layer

Handles REST API requests and responses.

Components:

- Controllers
 - Swagger Documentation
-

Business Logic Layer

Implements business logic and validation.

Components:

- Services
 - Report Generation Logic
-

Data Access Layer

Performs database operations.

Components:

- Spring Data Repositories
 - MongoTemplate
-

Database Layer

Stores application data in MongoDB.

Components:

- MongoDB Collections
 - Indexes
-

Project Structure

AeroPassport

controller

SupplierController

PartController

ModuleController

ReportController

DatabaseAdminController

service

SupplierService

PartService

ModuleService

ReportService

repository

SupplierRepository

PartRepository

PartPassportRepository

ModuleRepository

model

Supplier

Part

PartPassport

Module

Inspection

Event

dto

Exception

config

AeroPassportApplication.java

Database Design

MongoDB is used as the primary database.

The application consists of four collections.

Collection	Purpose
suppliers	Stores supplier details
parts	Stores hardware components
part_passports	Stores lifecycle history and inspections
modules	Stores subsystem assemblies

MongoDB Data Modeling Strategy

The project uses a hybrid document modeling approach.

Referencing

Used when entities have independent lifecycles.

Examples:

Supplier → Parts (One-to-Many)

Part → Passport (One-to-One)

Module ↔ Parts (Many-to-Many)

Passport → Inspections/Events (Embedded One-to-Many)

Embedding

Used for closely related historical information.

Examples:

Passport

 Inspections

 Events

MongoDB Indexing Strategy

To improve query performance, indexes are implemented.

Collection	Indexed Field	Type
parts	supplierId	Single Field
parts	batchNumber	Single Field
part_passports	partId	Unique
modules	partIds	Multikey

REST API Design

The system exposes RESTful APIs.

Base URL

http://localhost:8080/api

Resource	Endpoint	CRUD
Suppliers	/suppliers	POST, GET, PUT, DELETE
Parts	/parts	POST, GET, PUT, DELETE
Modules	/modules	POST, GET, PUT, DELETE

Endpoint	Purpose
/api/parts/{id}/genealogy	Hardware Genealogy Report
/api/admin/statistics	Dashboard Statistics
/api/admin/reset	Reset Database

Logging Strategy

Spring Boot logging is used to record major lifecycle actions.

Examples:

INFO Supplier created: SUP001

INFO Part registered: PAR001

INFO Passport generated: PAS001

INFO Event appended: Received

INFO Genealogy report generated.

Purpose:

- For console-based execution.
- Improve auditability.

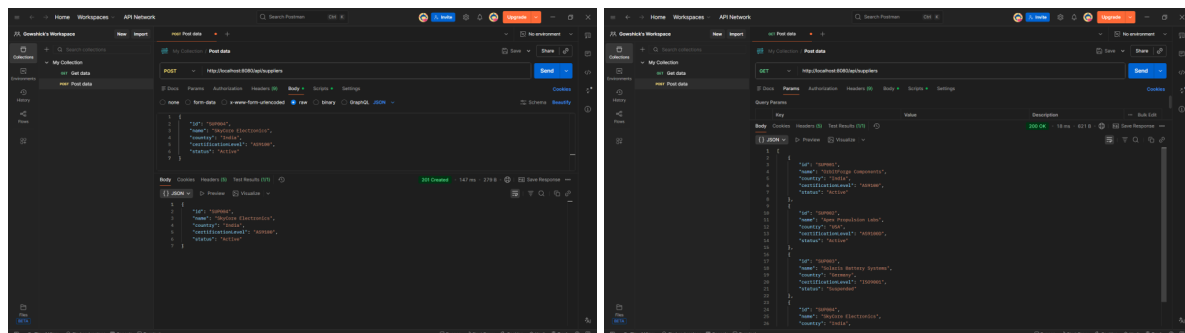
Verification and Testing Strategy

Manual verification was performed using Postman, Swagger UI, MongoDB Compass, and mongosh to validate CRUD operations, aggregation reports, indexes, embedded documents, and collection management.

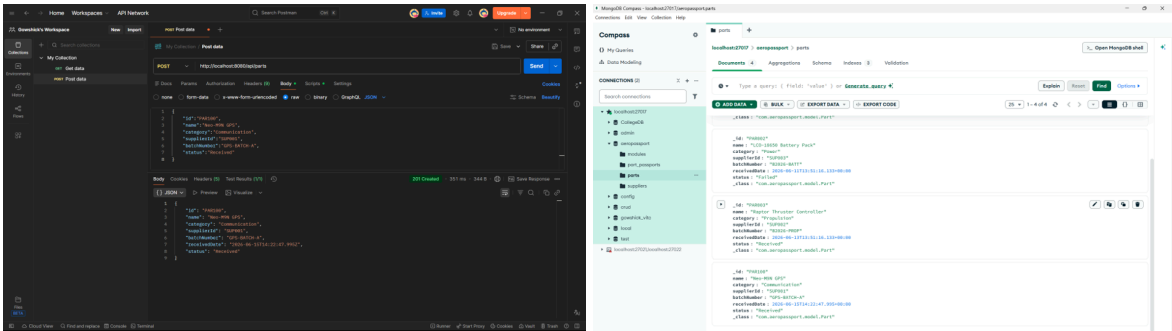
Implementation Screenshots

Note: Only a few screenshots have been added here for reference, complete screenshots are uploaded as a separate file.

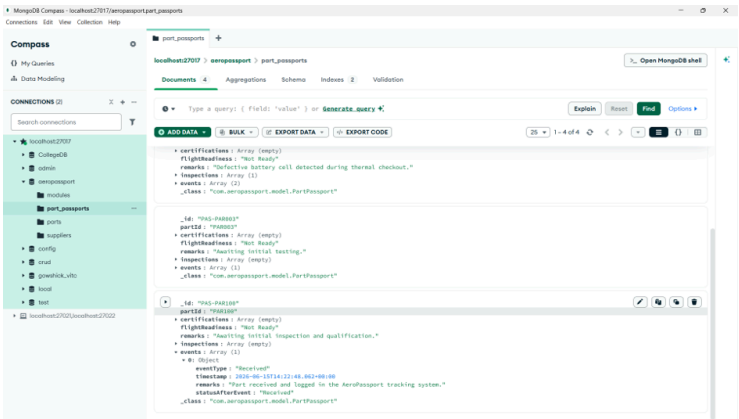
Supplier Creation and Verification:



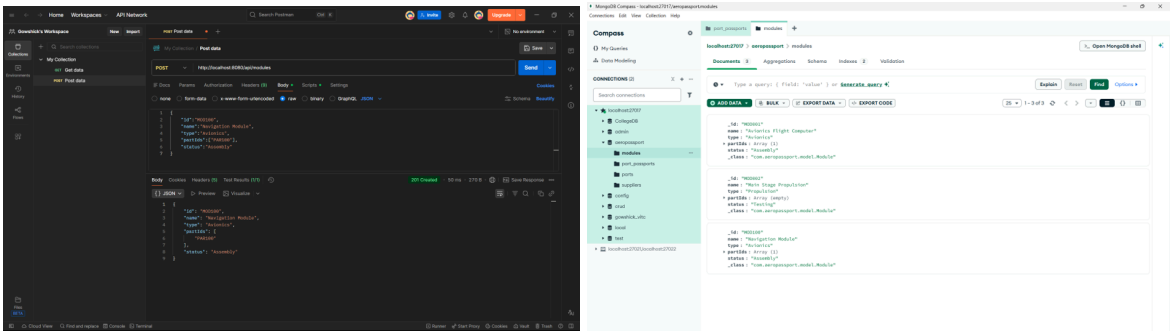
Parts Creation and Verification:



Automatic Passport Creation Verification:



Module Creation and Verification:



Project Outcome

AeroPassport was successfully developed as a MongoDB-based backend application for aerospace hardware traceability. The system supports supplier management, hardware registration, digital part passports, subsystem modules, lifecycle tracking, and automated event logging. It demonstrates CRUD operations, relationship mapping, embedded documents, indexing, aggregation pipelines, and custom MongoTemplate queries.

Project Summary

AeroPassport is a lightweight flight hardware traceability system that demonstrates MongoDB's document-oriented capabilities through a practical aerospace use case. Using Spring Boot, Spring Data MongoDB, REST APIs, aggregation pipelines, embedded documents, and indexing, the application efficiently manages interconnected hardware data while remaining scalable and maintainable.

Conclusion

AeroPassport demonstrates the practical application of MongoDB in solving real-world hardware traceability problems through document modeling, embedded documents, references, indexing, aggregation, and RESTful APIs. The project fulfills all certification requirements while showcasing modern backend development practices and MongoDB's strengths in managing interconnected lifecycle data.
